STA 111 Bulkhead Cabin Fans (S.O.O. 6378) – General Data

1. General

Provisions have been made for the installation of two electrically-driven fans in the cabin, on the aft face of the STA 111 bulkhead. The fans can be used to direct air around the passenger cabin for cooling purposes. The fans are located on each side of the cockpit door. A formed sheet metal stand-off assembly is riveted to the bulkhead. Each fan is attached to the stand-off assembly with three (3) machine screws. A Lone Star 122253–10A EMI filter is installed in the circuit. The filter is mounted on the pulley bracket box, installed on the forward side of the STA 111 bulkhead on the left side just above and behind the Pilot.

The fans are operated by a CABIN FANS STA 111 switch located in the flight compartment, above and behind the Pilot on the STA 111 bulkhead adjacent the EMI filter. Each fan has an integral OFF–LOW SPEED–HIGH SPEED switch within the base of the fan assembly, to allow for passenger adjustment.

28 Volt DC power is provided from the right bus through a 15 amp CABIN FANS (B14) circuit breaker on the main circuit breaker panel (refer to 24-00-00, Electrical Power – General Data).

2. Adjustment/Test

A. Operational Test

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Select CABIN FANS STA 111 switch to ON.

3) Move the integral switch located at the base of each fan through the OFF, LOW SPEED and HIGH SPEED positions, observing the operation at each setting.

Note The fan blades and motors should be smooth running without irregular noise, rattling, or vibration.

4) Select CABIN FANS STA 111 switch to OFF.

5) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

3. Removal/Installation

A. Remove STA 111 Bulkhead Cabin Fans

1) Ensure aircraft power is OFF. (Refer to 24-00-00, External Power – General Data).

2) Pull the CABIN FANS (B14) circuit breaker from the main circuit breaker panel and fit safety clip.

3) Remove three (3) mounting screws and washers on the base of each fan.

4) Before disconnecting the electrical leads from the terminal on the fan assembly, discharge the EMI filter through a +A terminal either by:

a) Connecting the +A terminal to an aircraft ground through a 150 ohm, 7.5 watt resistor for at least 30 seconds; or,

b) Connecting a wire to an aircraft ground and discharging the EMI filter by touching the other end of the wire to the +A terminal.

Note Do not re-engage circuit breaker until leads are reconnected to fan assembly terminals.

Note If leads need to be subsequently removed for further maintenance, repeat discharge procedures in Para 3.A.4) prior to lead disconnection.

5) Carefully pull the fan from the stand-off assembly, and disconnect the electrical wiring.

6) Remove the fan assembly from the aircraft.

B. Install STA 111 Bulkhead Cabin Fans

1) Ensure aircraft power is OFF. (Refer to 24-00-00, External Power – General Data).

2) Ensure CABIN FANS (B14) circuit breaker is pulled and fitted with safety clip.

3) Examine the fan assembly for damage, replace if necessary.

4) Place the fan assembly near the stand-off assembly and connect the electrical wiring.

Note If leads need to be subsequently removed for further maintenance, repeat discharge procedures in Para 3.A.4), prior to lead disconnection.

5) Install three (3) mounting screws and washers. Ensure the fan assembly is secure on the stand-off assembly.

6) Remove the safety clip and close CABIN FANS (B14) circuit breaker on the main circuit breaker panel.

7) Do an operational test as per Para 2.A.

4. Servicing

A. Fan Failure

Note The fan is not repairable. If a fan is unserviceable, it must be replaced.

1) Ensure aircraft power is OFF. (Refer to 24-00-00, External Power – General Data).

2) Remove the three (3) screws fastening the fan base to the stand-off assembly.

3) Pull the wiring out and disconnect the wiring splices.

4) Replace the fan assembly with a new part.

5) Connect the new fan to the wiring splices and re-install screws.

5. Troubleshooting

A. Fan Fault

1) Ensure CABIN FANS (B14) circuit breaker is closed.

2) Ensure the integral switch at the bottom of each fan is in the LOW or HIGH position.

3) Ensure the CABIN FANS STA 111 switch is ON.

4) If troubleshooting at a wire level is required, refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.